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An Allometric Study of Hippocampal Components

A Comparative Study of the Brains of the European Hedgehog *(Erinaceus europaeus)*, the Tree Shrew *(Tupaia glis)*, and the Marmoset Monkey *(Callithrix jacchus)*

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Key Words. Allometric investigation · *Callithrix* · *Erinaceus* · European hedgehog · Hippocampus · Marmoset · Tree shrew · *Tupaia*

Abstract. Brains of western European hedgehogs, common tree shrews and common marmosets were perfused and stained according to Timm's method. The volumes of hippocampal components were determined with semiautomated quantitative techniques and compared by allometric methods. The results support previous findings that the size of the hippocampus in marmosets and tree shrews is about twice that of the hippocampus of the European hedgehogs of the same body weight. Three hippocampal substructures account for most of this progression. These are the hilus, the deep zone of the molecular layer of the area dentata, and CA1 (especially the cellular layer) of Ammon's horn.

Introduction

The hippocampus is an allometrically progressive structure. That is, if the hippocampal volumes of basal insectivores and primates of the same body weights are compared, the larger volumes are found in primates [Stephan, 1975]. From surface measurements, Stephan and Manolescu [1980] concluded that the CA1 [cornu ammonis 1; Lorente de Nó, 1934] component of the hippocampus contributes most to this progression. A comparison of the microscopic appearance of the hippocampus of a basal insectivore and a primate (fig. 1) also gives the impression that the CA1 is proportionally larger in the latter. In order to determine the degree to which the various components contribute to the progression of the hippocampus in mammals, a more rigorous volumetric analysis of the subdivisions of area dentata, Ammon's horn and subiculum has been performed.

For a comparative study, we chose the brains of the western European hedgehog *(Erinaceus europaeus)*, the common tree shrew *(Tupaia glis)*, and the common marmoset *(Callithrix jacchus)*. Hedgehogs belong to the order Insectivora, which shares many characteristics with the early insec-

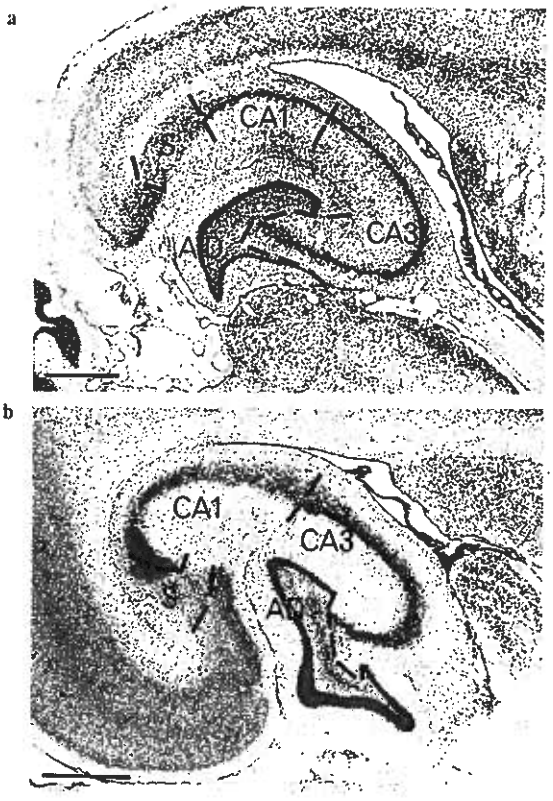


Fig. 1. Horizontal Nissl-stained sections of the western European hedgehog, *E. europaeus* **(a)** and the common marmoset, *C. jacchus* **(b)**, showing the cytoarchitectonic differentiations of the area dentata (AD), fields CA3 and CA1 of Ammon's horn, and the subiculum (S). Scale bars = 1 mm.

tivorous mammals that comprised the stock from which the mammalian orders evolved [Romer, 1966; Starck, 1978]. Since the neocortex has undergone considerable evolution in size among mammals, the dimensions of the brains of those insectivores which display the highest degree of evolutionary conservatism (i.e., in which the neocortex is least developed) can be used to construct base lines for inter-order comparisons. Of the large number of living insectivore species, 12 species (which include the western European hedgehog) fall into this category, which has been designated 'basal insectivore' by Stephan [1967]. The second animal used in this study, the common tree shrew, has previously been grouped with both the primates and the insectivores [for review, see Luckett, 1980]. More recently, the common tree shrews have been given their own order, Scandentia [Corbet and Hill, 1980]. Like Starck [1965] and others, we regard them here as descendants of a separately derived line that shared a common pool of ancestors with the primates. The third animal in our study, the common marmoset, is a true simian belonging to the suborder Platyrrhini or New World monkeys. It is generally believed to have one of the most conservative primate brains [Thenius, 1967].

In rodents and insectivores, Timm's heavy-metal sulfide stain [Timm, 1958] has been used successfully to define the boundaries of the subdivisions of the hippocampus. This has permitted us to determine the volumes of the hippocampal components [West et al., 1978, 1984]. The well-defined borders of the regions of differential staining seen with Timm's stain appear to be related to differences in the heavy-metal chemistry of the highly segregated terminal fields of the intrinsic and extrinsic fiber systems of this part of the cortex. These regions correspond, to a large degree, to the divisions of the hippocampus seen in both hodological and architectonic studies of other species [Haug, 1974; Zimmer and Haug, 1978; West et al., 1984]. In order to identify trends in the structural evolution of the mammalian

Table I. Weight and volume data of animals studied

	Body weight g	Telencephalic volume mm³	Brain weight mg
Hedgehog, mean[a]	860	1,931	
H_5	640	2,140	
H_9	500	2,032	
H_{11}	500	1,514	
H_{15}	530	1,650	
Marmoset, mean[a]	280		7,600
M_1	304		7,060
M_2	215		7,000
Tree shrew, mean[a]	170		3,200
T_1	296		3,330
T_2	186		3,310

[a] Data from Stephan et al. [1981].

hippocampus, the volumes of the subdivisions of the Timm-stained hippocampus of *Tupaia* and *Callithrix* were compared to their positional homologues in the western European hedgehog *(E. europaeus).*

Material and Methods

For volumetric measurements, we used the brains of 4 hedgehogs, 2 tree shrews, and 2 marmoset monkeys (table I). The material used to represent the hedgehog has been described in detail previously [West et al., 1984].

Preparation of Brains

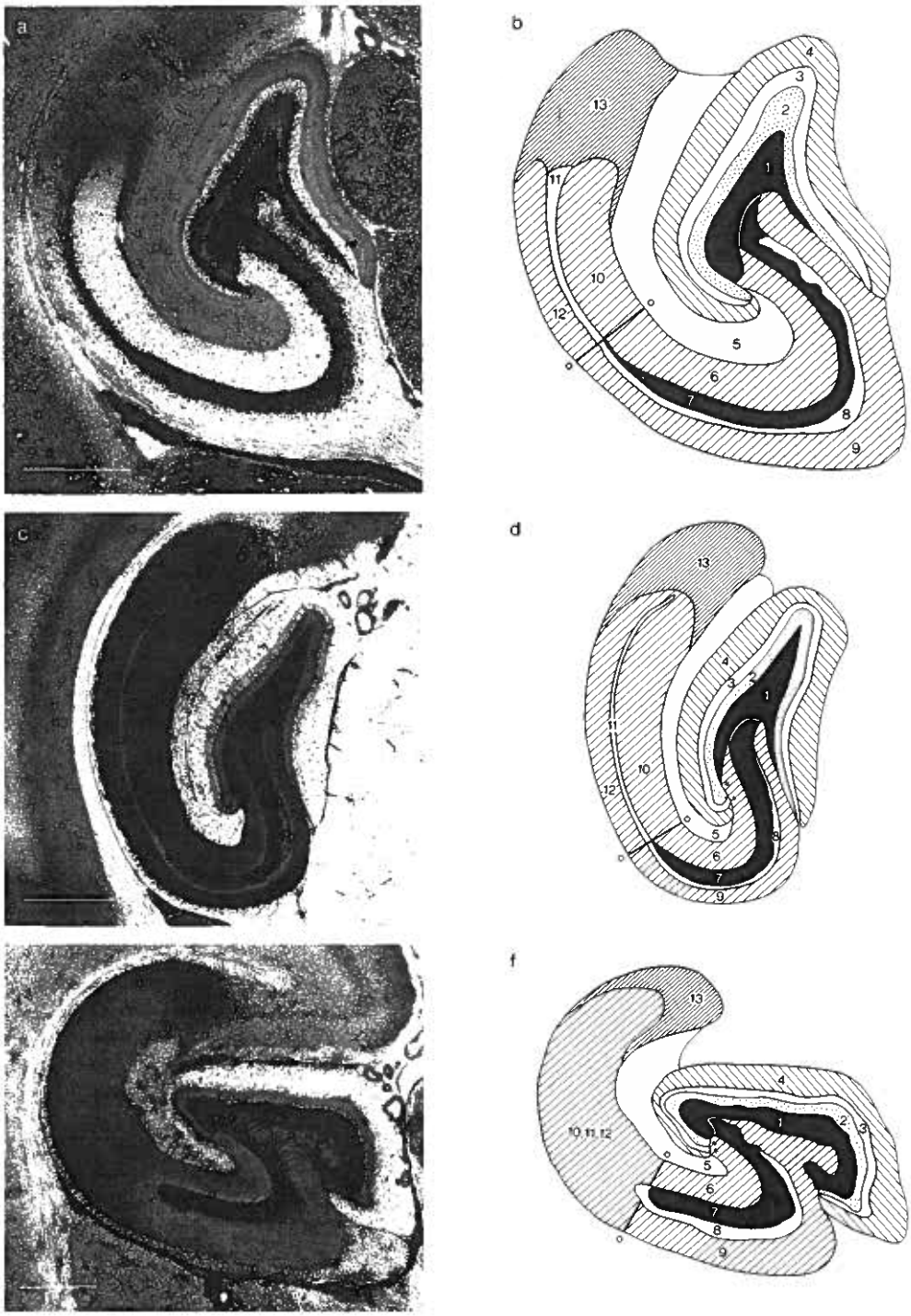
The histological material was prepared according to a modification of the Timm technique that utilizes a 0.1% Na_2S perfusion and cryostat sections [Danscher, 1981]. The forebrain of one of the *Tupaia* and *Callithrix* specimens was sectioned in the horizontal plane while that of the other specimens was sectioned in the frontal plane. The 4 *Erinaceus* brains were sectioned horizontally. Four series of alternate sections (30 μm thick) with varying development times were prepared for each brain. Some of the sections were counterstained with thionine to facilitate the correlation of the staining pattern with the cytoarchitecture. The histological procedure was essentially the same as that used with the hedgehogs referred to above.

Calculation of Volumes of Hippocampal Substructures

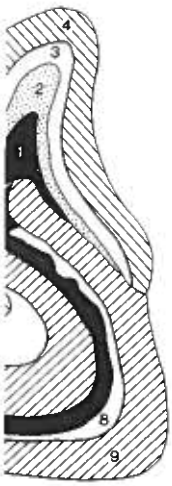
From the Timm-stained series, every 20th section was photographed, and the borders of hippocampal areas and layers were marked as shown in figure 2. The technique used for determining the volumes of the components of the hippocampi was the same as that used in a previous study of rats [West et al., 1978], except that the borders were transformed to a series of x-, y-coordinates with a computer-controlled (PDP-11, DEC) Vidissector camera (ITT).

Allometric Method

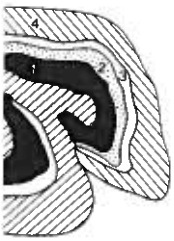
The volumes of the components of the hippocampus were compared in isoponderous examples of each species by the allometric method, in which the relationship between the body weight and the volume of the brain (or its components) is determined by calculating regression lines. The slopes of the regression lines indicate the rate at which the brain or its com-



2







ponents enlarge with increasing body weights. For the hippocampus of insectivores and primates, this regression line is expressed by the equation $\log \bar{y} = 0.57 \times \log \bar{x} + b$ [Stephan and Manolescu, 1980], in which $\bar{y}$ is the mean volume of the structure under consideration, $\bar{x}$ the mean body weight of the species, and b the ordinate point of the line which is specific for each structure. Calculations were performed according to the methods of Stephan as described in detail by Stephan et al. [1981] and Frahm et al. [1982]. Briefly:

(1) The volume data from all brains were 'standardized' by multiplying the volume of each hippocampal component in the tree shrews and marmosets by the ratio of the mean brain weight of the species [obtained from Stephan et al., 1981] to the actual brain weight of the individual specimens. For the European hedgehogs, we used the ratio of the average telencephalic volume to the individual telencephalic volume since brain weight data were unavailable. This correction is necessary in order to make statements about interspecies relationships on the basis of data from a small number of specimens. It is permissible, since there is no evidence of directed changes in the proportions of the components of adult brains related to the size of the brain [Stephan et al., 1970].

(2) For each hippocampal component, the b value was determined by using the mean of the standardized volumes of the specific component in *Erinaceus* as $\bar{y}$ in the allometric equation.

(3) As a first step in comparing the volumes of the hippocampal components in *Tupaia* and *Callithrix* with those in hedgehogs, 'reference volumes', corresponding to the volumes of hippocampal components in hypothetical hedgehogs with the same body weights as tree shrews and marmosets were calculated. To accomplish this, the average body weights (b. w.) of the species were inserted into the allometric equation. For example the reference volume (ref. T) of the fascia dentata (f. d.) in common tree shrews would be expressed by the following relationship:

$$\log (\text{ref. T}) = 0.57 \cdot \log (\text{b. w. T.}) + b\ (\text{f. d.}).$$

(4) The second step was to compare the standardized component volumes for each individual with the reference volume. The ratio of these values, multiplied by 100, was called the size index (SI). For example, the SI of the fascia dentata in common tree shrews would be expressed by the following relationship:

$$\text{SI} = \frac{\text{standardized volume of f. d. in an individual tree shrew} \cdot 100}{\text{reference volume of f. d. for tree shrew}}.$$

(5) Finally, the mean SI was calculated from the individual SIs. If, for example, one of the hippocampal components of the tree shrew has an SI of 200 this means that the volume of this component is twice as large in the *Tupaia* brain as it is in the brain of a hypothetical hedgehog with the same body weight. An increase of this type (i.e., in which the SI is greater than 100) is referred to as 'progression'.

Fig. 2. Horizontal sections of the hippocampus of the western European hedgehog, *E. europaeus* **(a, b)**, the common tree shrew, *T. glis* **(c, d)**, and the common marmoset, *C. jacchus* **(e, f)**. On the left are representative Timm-stained sections, and on the right illustrations of the boundaries of the components that were measured. The numbers refer to specific hippocampal components: 1 = hilus fasciae dentatae; 2 = stratum granulosum; 3 = deep zone of stratum moleculare fasciae dentatae; 4 = superficial zones of stratum moleculare fasciae dentatae; 5 = stratum eumoleculare-lacunosum of Ammon's horn and subiculum; 6 = stratum radiatum (CA3); 7 = mossy-fiber zone; 8 = stratum pyramidale (CA3); 9 = stratum oriens (CA3); 10 = stratum radiatum (CA1); 11 = stratum pyramidale (CA1); 12 = stratum oriens (CA1); 13 = deep zone of the subiculum. The lines between the open circles (O—O) represent the borders between regio inferior (CA3) and regio superior (CA1). Asterisks indicate the border between stratum moleculare of area dentata (3, 4) and that of regio inferior (CA3) (5). Scale bars = 1 mm.

Results and Discussion

Description of Components and Their Boundaries

The organization of the components of the Timm-stained hippocampus in tree shrews and in marmosets is similar to that seen in rats, mice and hedgehogs. As a consequence, the boundaries used for the volumetric determinations made in this study are similar to those described in previous studies [West et al., 1978, 1980, 1984]. The major subdivisions (i.e., area dentata, Ammon's horn and subiculm) are all discernible and the boundaries of their subdivisions readily defined. Differences in the staining intensities of the various components can be appreciated by comparing the positional homologues in the different species shown in figures 2a, c, e. The following briefly describes the components that were compared and the features of the hippocampus that differed significantly from other species.

Area dentata. In tree shrews and marmosets, the transversely sectioned hilus fasciae dentatae (hilus) appears as a densely stained crescent-shaped core of the area dentata (fig. 2d, f: component 1). It stains intensely black and shows no regional differentiation along the septotemporal axis. In both species it lies deep to the well-defined granular cell layer (fig. 2d, f: component 2). Within the molecular layer of the area dentata, variations were observed in the laminae oriented perpendicular to the radial plane. In both the tree shrew and the marmoset the deep zone of the molecular layer (fig. 2d, f: component 3) was, for the most part, darker than the superficial zones (fig. 2d, f: component 4). In the tree shrew, however, there was a tendency for the relative intensities of the deep and superficial zones to be reversed near the temporal pole. In the tree shrew, 2 subzones could also be identified within the deep zone and 3 subzones within the superficial zone at temporal levels. The differences in the staining of these subzones was subtle, less dramatic than those observed in the superficial zone of the albino rat [West et al., 1978], and precluded a separate volumetric analysis. In marmosets, the light layer seen immediately subjacent to the granule cell layer was more pronounced in the temporal direction and was included within the boundary of the hilus. At septal levels (i.e., parts of the hippocampus closest to the septum) significant accumulations of dark granules, similar to those seen in the hilus and the mossy-fiber zone, were also found between the granule cell and molecular layers of the hidden or suprapyramidal blade (fig. 2e, between the ends of components 2 and 3 that lie to the left). These accumulations were included within the boundaries of the deep zone of the molecular layer. The border between the molecular layer of the area dentata and that of Ammon's horn was defined by a line drawn from the tip of the granule cell layer in the suprapyramidal blade of the fascia dentata (fig. 2d, f: upper asterisk) to the deepest part of the hippocampal fissure (fig. 2d, f: lower asterisk).

Ammon's horn. Ammon's horn could be divided into a regio inferior (CA3) and regio superior (CA1) in all 3 species on the basis of the mossy fibers (fig. 2b, d, f: component 7) that stain intensely with the Timm technique. In neither the Timm nor the Nissl preparations was it possible to define a separate CA2. As a consequence,

Table II. The standardized[a] volumes of the hippocampal components of four western European hedgehogs H_5, H_9, H_{11}, H_{15} and their means

	Individual standardized volumes, mm³				Mean of standardized volumes, mm³
	H_5	H_9	H_{11}	H_{15}	
Total hippocampus	200.80	157.30	204.72	204.90	191.98
Area dentata	57.56	44.84	52.64	61.20	54.07
Hilus	9.92	7.74	8.62	10.60	9.22
Fascia dentata	47.64	37.10	44.02	50.60	44.85
Granular layer	9.06	8.90	8.82	10.16	9.24
ML	38.58	28.20	35.20	40.44	35.61
deep ML	8.02	5.80	9.08	7.96	7.72
middle/superficial ML	30.56	22.40	27.24	31.36	27.89
Subiculum (without SEL)	23.04	24.22	24.22	29.40	26.72
Ammon's horn (without SEL)	98.46	71.76	99.28	90.64	90.04
CA1 (without SEL)	30.40	16.02	31.62	32.56	27.65
CA3 (without SEL)	68.06	55.74	67.66	58.09	62.39
SEL, Ammon's horn + subiculum	21.74	16.54	22.58	23.74	21.15

These volumes were obtained by multiplying the volumes of the components of each specimen by the ratio of the mean volume of the telencephalon of the species (table I) to the mean volume of the telencephalon of the four hedgehog specimens used in this study. ML = Molecular layer. SEL = Stratum eumoleculare-lacunosum.

'regio superior' is used synonymously with CA1 and 'regio inferior' synonymously with CA3. The regio inferior of *Erinaceus* included the main and the tip zone of Ammon's horn, as described previously [West et al., 1984]. For quantitative purposes the boundary between the regio inferior and the regio superior was drawn at the tip of the mossy-fiber zone along the radial axis (fig. 2b, d, f: double circles). In the common tree shrew (fig. 2c, d) this boundary proved to be essential, in that the lack of regional differences along the dentato-subicular axis of Ammon's horn precluded the definition of components in regio superior and regio inferior on the basis of the morphology of the other layers. However, there were significant differences in the organization of these layers in the regio superior and regio inferior of the common marmoset (fig. 2e, f). This was primarily due to the presence of the pyramidal cells at various positions along the radial axis of the regio superior rather than in a single layer. It was, consequently, not possible to define a separate stratum oriens (12), stratum pyramidale (11) or stratum radiatum (10) in the regio superior of the marmoset. Beside these dramatic differences in the regio superior of the marmoset (which required the use of a collective zone that was comparable to a combination of strata oriens, pyramidale, and radiatum; fig. 2e, f: components 10, 11, 12) the quantified zones were essentially the same as those seen in the hedge-

Table III. Volumes (mm³) and SI of hippocampal components

	Species						
	hedgehog			tree shrew			
	mean of standardized volumes	reference volume		individual standardized volumes		SI	
		tree shrew	marmoset	T_1	T_2	T_1	T_2
Total hippocampus	191.98	76.27	101.36	137.17	153.72	180	202
Area dentata	54.07	21.45	28.50	40.30	39.51	188	184
Hilus	9.22	3.66	4.86	7.26	7.41	198	202
Fascia dentata	44.85	17.78	23.65	33.04	32.10	186	181
Granular layer	9.24	3.67	4.87	6.17	5.65	168	154
ML	35.61	14.14	18.79	26.87	26.45	190	187
deep ML	7.72	3.06	4.07	7.05	7.54	230	246
middle + superficial ML	27.89	11.08	14.72	19.82	19.33	179	174
Subiculum (without SEL)	26.72	10.60	14.09	19.08	15.02	180	142
Ammon's horn (without SEL)	90.04	35.76	47.52	62.09	82.45	174	231
CA1 (without SEL)	27.65	10.97	14.58	33.10	41.28	302	376
CA3 (without SEL)	62.39	24.74	32.88	28.99	41.17	117	166
SEL, Ammon's horn + subiculum	21.15	8.40	11.17	15.70	16.74	187	199

T_1, T_2 = Tree shrew specimens 1 and 2, respectively; M_1, M_2 = marmosets 1 and 2.

hog (fig. 2a, b). The lighter-staining substratum eumoleculare-lacunosum (SEL) was distinguishable in both regions of Ammon's horn in both *Tupaia* and *Callithrix* (fig. 2d, f: component 5), and appeared to be continuous with the most superficial zone of the subiculum.

Subiculum. In both *Tupaia* and *Callithrix*, the Timm-stained subiculum appears to have two components. The superficial pale zone is continuous with the similarly stained stratum eumoleculare-lacunosum of Ammon's horn (5) and is distinct from the darkly stained deep zone (13) which appears to embrace the subicular end of the deeper layers of Ammon's horn, as it does in rat, mouse and hedgehog. Because no natural border was apparent between the superficial zone of the subiculum and the SEL of Ammon's horn, these components were considered as one. This approach resulted in a component that included the 'molecular layers' of the regio inferior, regio superior and subiculum (5), and was comparable to that defined in earlier work on the hedgehog [West et al., 1984].

l zed	SI		Species				Mean SI	
			marmoset				tree shrew	marmoset
			individual standardized volumes		SI			
T_2	T_1	T_2	M_1	M_2	M_1	M_2		
153.72	180	202	186.00	180.55	184	178	191	181
39.51	188	184	45.17	50.13	158	176	186	167
7.41	198	202	12.62	11.25	260	231	200	246
32.10	186	181	32.55	38.88	138	164	184	151
5.65	168	154	6.37	6.75	131	139	161	135
26.45	190	187	26.18	32.13	139	171	189	155
7.54	230	246	6.85	8.79	168	216	238	192
19.33	179	174	19.33	23.34	131	159	177	145
15.02	180	142	14.12	14.94	100	106	161	103
82.45	174	231	106.52	96.61	224	203	203	214
41.28	302	376	56.56	53.07	388	364	339	376
41.17	117	166	49.96	43.54	152	132	142	142
16.74	187	199	20.19	18.87	181	169	193	175

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Allometric Comparisons

The volumes and size indices of the hippocampal components analyzed here are given in tables II and III and in figure 3. The hedgehog data are considered to be representative of the basal insectivores, since Stephan and Manolescu [1980] reported that the allometric indices of the main subdivisions of the hedgehog hippocampus are at intermediate levels in the range of values for basal insectivores.

Hippocampus. In *Callithrix* and *Tupaia*, the hippocampus, like the brain, is (allometrically) larger than in *Erinaceus* [Stephan and Manolescu, 1980]. Our hippocampal size indices (fig. 3a; table III) are similar to those of Stephan and Manolescu, which were based on comparisons with a group of basal insectivores. We found, however, that the SI for *Callithrix* was a little smaller than that for *Tupaia*. Stephan and Manolescu reported the opposite. This difference may be due to the small *Callithrix* specimens used in this study or due to the fact that we excluded the pre- and supracommissural hippocampi and the alveus/fimbria/fornix system from our comparisons. These components subjectively appear to be more prominent in the marmoset than in the tree shrew.

Area dentata. The area dentata is 1.86 times larger in the tree shrew and 1.67 times larger in the marmoset than in the hedgehog. This volume increase is primarily due to an increase in the hilus (fig. 3b), especially in *Callithrix*. In the intrahippocampal chain of projections from the area dentata to CA3 to CA1 to the subiculum [Swanson et al., 1982; Schwerdtfeger, 1984] the hilus is interposed between the granular cells of the fascia dentata and the neurons of CA3. Hilar cells have been reported to be the sole source of the commissural-associational projection which terminates in the deep zone of the molecular layer on granule cell dendrites [West et al., 1979; Laurberg and Sørensen, 1981]. This is one of the few projections that runs against the main follow of information along the dentata-subicular axis. Recently, it has been found that most of the hilar cells are of the non-pyramidal type [Köhler, 1983], possibly interneurons. There-

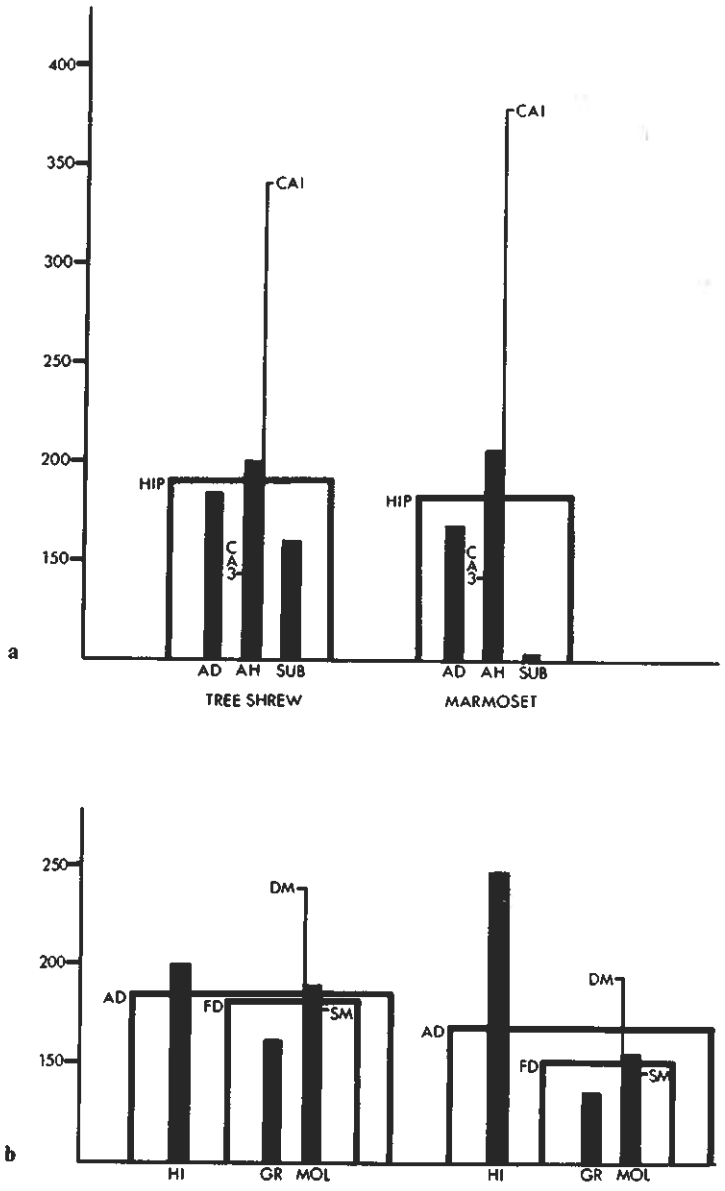


Fig. 3. SI (ordinate) of main hippocampal divisions **(a)** and substructures of the area dentata **(b)** in the brains of tree shrews and marmosets. AD = Area dentata; AH = Ammon's horn; DM = deep zone of molecular layer of fascia dentata; FD = fascia dentata; GR = granular layer of fascia dentata; HI = hilus of area dentata; HIP = hippocampus; MOL = molecular layer of fascia dentata; SM = middle and superficial zones of molecular layer of fascia dentata; SUB = deep subiculum.

fore the hilar projection to the molecular layer is most likely a recurrent inhibitory circuit. The progression observed in the hilus suggests that the role of this feedback loop in hippocampal function may be more significant in more advanced mammals than in insectivores.

While the entire molecular layer of the fascia dentata is progressive in both *Tupaia* and *Callithrix* (SI = 189 and 155, re-

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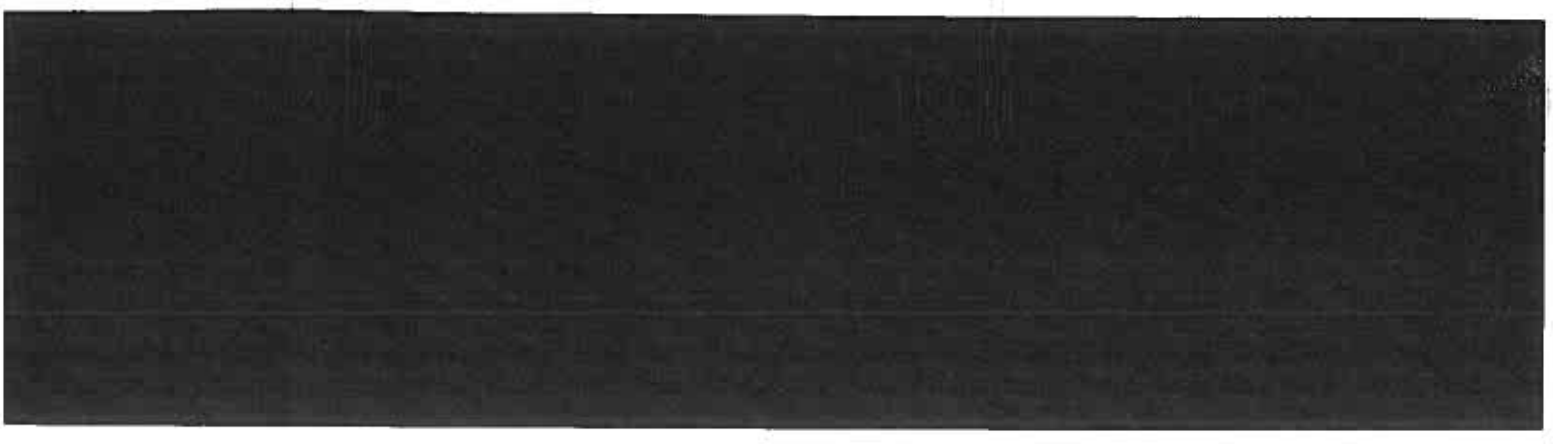
spectively), most of the progression can be attributed to the deep zone (fig. 2d, f; component 3; SI = 238 and 192, respectively). This zone is the terminal field of the commissural-association projection from the hilus which, as stated above, has the largest SI of all the components that were measured. A similar correlation between the size of the hilus and the deep zone of the molecular layer has been observed in two rodent species [West and Andersen, 1980]. Although an analysis of the coevolution of converging neuronal systems is beyond the scope of the present study it may be of value to future studies to point out that the other major fiber projection to the molecular layer, the perforant path, which terminates on the same cells in more superficial layers (fig. 2d, f: component 4), is also progressive, though to a lesser degree, and that the progression in the size of one terminal field has not occurred at the expense of the other.

Subiculum. The subiculum is larger in *Tupaia* and *Callithrix* than in the hedgehog, but shows only a moderate progression that is clearly lower than that for the total hippocampus. The subiculum is a nodal point for afferent and efferent extrahippocampal projections [Swanson et al., 1982]. The proportionally smaller subiculum of the more advanced forms is surprising in that it was expected that a structure that connects the hippocampus to other parts of the brain would reflect the progression seen in the other major subdivision of the hippocampus.

The smaller degree of progression measured in the subiculum of *Tupaia* and *Callithrix* may be explained in one of the following ways: Either the progression measured in the rest of the hippocampus is primarily related to a progression in elements involved in intrahippocampal activities, and is therefore not reflected in components that connect the hippocampus to other structures, or the role of the subiculum as a nodal point for extrahippocampal projections is now shared with other structures that were not measured. Resolution of the problems related to differential progression of connected neural structures will require volumetric data from other parts of the brain known to be related to the hippocampus and additional comparative studies of the connections of these areas.

Ammon's horn. The measurements made within the Ammon's horn suggest that most of the progression of the hippocampus is related to an increase in intrahippocampal circuitry. The size indices of the Ammon's horn indicate that it is the most progressive of the major hippocampal subdivisions that were studied. Three findings are noteworthy: (1) In *Tupaia* and *Callithrix* the regio superior (CA1), as defined in Timm preparations, is larger than regio inferior (CA3; see table II), whereas the opposite is true in the hedgehog: (2) the regio superior (CA1) shows the highest progression of all hippocampal components (table II; fig. 3a); and (3) it is obvious in Niss preparations of the *Callithrix* hippocampus (compare fig. 1a, b) that the pyramidal cells of CA1 are also found, to a large extent, in the stratum oriens [as they are in other simians; Stephan and Manolescu, 1980; Stephan, 1983].

The hippocampus is involved in specific aspects of memory and other higher brain functions [O'Keefe and Nadel, 1978; Olton et al., 1980; Winocur, 1980; Swanson et al., 1982; Braitenberg and Schüz,



1983; Schwerdtfeger, 1984]. According to the 'principle of the proper mass' formulated by Jerison [1973], 'the mass of tissue controlling a particular function is appropriate to the amount of information processing involved in performing the function'. The progression in the hippocampus of *Tupaia* and *Callithrix* demonstrated here indicates that in these species the hippocampus, and in particular CA1, can process more information relative to body size than its homologue in insectivores. Field CA1 receives primarily intrahippocampal afferents [for reviews, see Swanson et al., 1982; Schwerdtfeger, 1984]. Since field CA1 appears to be primarily involved in the integration and/or modulation of intrinsic (association) impulses, the large amount of progression in this region suggests that the relative capacity for intra-hippocampal associations is greater in the hippocampus of the two more advanced species than in that of the basal insectivore.

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